🌀 M003 Extension — The Loop Spiral (Intermediate)

Topic: One loop that **turns** (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Intermediate, after the Loop Staircase) · **Time:** about 40 minutes

The **staircase** took the **same** step every pass, so it grew in a **straight diagonal**. A **spiral** also **turns** every pass — same loop, but the direction **changes** — so the squares wrap into a tower. Type `SPIRAL` in the chat and watch it **climb** and **turn**.

```
x = 8
y = 0
z = 8
for step in range(12):
    fill GOLD from (x, y, z) to (x + 2, y, z + 2)
    y = y + 1
    if step % 4 == 0:
        x = x + 4
    elif step % 4 == 1:
        z = z + 4
    elif step % 4 == 2:
        x = x - 4
    else:
        z = z - 4
```

> 🔁 **Big idea:** `step % 4` counts **0, 1, 2, 3, 0, 1, 2, 3 …** — four directions on repeat. That is the **turn**. `y = y + 1` lifts each square higher. **Turn + rise = a spiral.**

	3 ↑ back **z − 4**	
2 ← left **x − 4**	**step % 4**	**0 → right** **x + 4**
	1 ↓ forward **z + 4**	

1 · Predict 🔮

Fill the **turn** for each step. `step % 4` repeats **0, 1, 2, 3**. Step 0 is done.

step	step % 4	turn (direction)
0	0	right (x + 4)
1		
2		
3		
4		
5		
6		
7		

`range(12)` **builds how many squares? Circle one:** 4 · 8 · 12

Which line makes the spiral go UP? Circle one: `y = y + 1` · `x = x + 4` · `step % 4`

Top-down, the 12 squares land on only 4 **spots. Why only 4? (1 sentence)**

2 · Spot the Bug 🐛

Bug A — no turn. A friend deleted the four `if` lines, so the direction never changes.

```
for step in range(12):
    fill GOLD from (x, y, z) to (x + 2, y, z + 2)
    y = y + 1
    x = x + 4
```

What do you build instead of a spiral? Circle one: a straight diagonal · a flat square · nothing

Bug B — no rise. The `y = y + 1` line is gone, so y stays **0**.

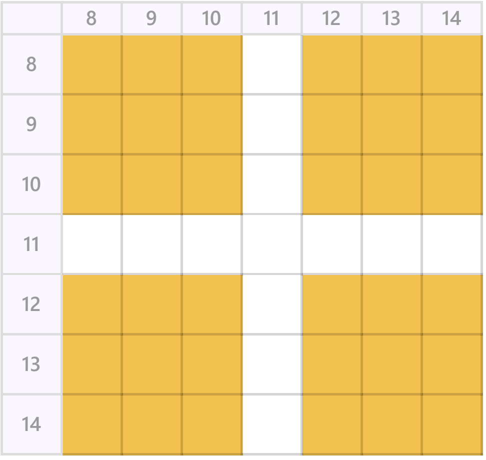
What happens to the tower? Circle one: it stays flat on the floor · it grows taller · it turns upside down

Write the missing line and say where it goes:

3 · Show Your Work 📸 🎥

Load the world. Stand on your home spot. Type `SPIRAL` and watch each square **turn** to a new corner and **climb** one block.

👁 Top-down view (looking DOWN): x → across, z ↓ deeper



Looking down, the squares sit on 4 corners — the loop turns right, forward, left, back, then repeats, climbing each time.

Modify challenge. Change **one** thing and **predict before you run** — for example `range(12)` → `range(20)` (taller), or the turn step `+ 4` → `+ 6` (wider). Write what will change, then build it.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what the loop does and how `step % 4` turns it.

2 · Your build. Point the camera at the spiral. Show it climbing and turning.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.

> Something new I learned was ______.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your **one** video to teacher on KakaoTalk.